LP581X
Family of I^2C LED driver IC's

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Introduction

The main purpose of this library is to simplify communication process between host MCU and slave device. Communication is handled by I^2C protocol. Please note that, this library has been created as hobby project and was not meant to be ideal.

Family of LP581X devices was designed by Texas Instruments. These ICs are relatively cheap devices that allow designers to easily integrate LEDs in devices, that are using I^2C protocol as their main internal communication channel. Library at the moment can cooperate with devices mentioned below:

LP5814: 4 - channel I2C interface RGBW LED driver with auto animation control.

LP5815: 3 - channel-Channel I2C Interface RGBW LED Driver with Instant Blinking and Auto Animation Control.

LP5816: 4 - channel I2C interface RGBW LED driver.

LP5817: 3 - channel I2C interface RGBW LED driver.

Mentioned above devices, can operate with voltages ranging from 2.5V to 5.5V. Logic pins are compatible with 1.8V, 3.3V and 5V standards. Devices are meant to sink current up to 51mA per channel. More information can be found in manufacturer datasheet.

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Library components

Constructor

Library has been based on OOP (Object Oriented Programming), therefore before usage of any component, user has to create an object with following scheme:

LP581X nameOfObject (LP581X_STD_ADR);

LP581X is the name of used device. Options include: **LP5814, LP5815, LP5816 and LP5817.**

nameOfObject is the name of created object, that user will refer to - for example **myLED.**

LP581X_STD_ADR is the standard I^2C address of the device. Options include: **LP5814_STD_ADR, LP5815_STD_ADR, LP5816_STD_ADR and LP5817_STD_ADR.**

enable()

In order to use the device, it has to be enabled. This function has to be called first, before any other command. Function does not utilise additional input arguments.

disable()

User can optionally disable the device, for the time being. Function does not utilise additional input arguments.

reset()

This operation resets all values of internal registers including enable register. Function does not utilise additional input arguments.

update()

Before any change made by user can take place, the special internal register has to be activated. User has to call update(), after other registers are set to selected values. Function does not utilise additional input arguments.

setMaxCurrent(bool maxCurrent = false)

User can select the maximal value of current, that can flow per one channel. Available options are 25.5mA and 51mA. Normally, maximal value of current is set to 25.5mA. This corresponds to setting function argument to false (or leaving argument blank). If the argument is set to true, maximal current value of 51mA will be selected.

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enableOutput(int firstOutput = false, int secondOutput = false, int third Output = false, fourthOutput = false[1])

User can select the amount of channels that will be used by the device. If no arguments are provided, all outputs will be enabled. User can use keywords: **OUT0, OUT1, OUT2 and OUT3**[1] as arguments. Order can be random.

enableFade(int firstOutput = false, int secondOutput = false, int third Output = false, fourthOutput = false[1])

User can utilise internally built fade function inside device, by setting on which output should it work. If no arguments are provided, all outputs will use fade function. User can use keywords: **OUT0, OUT1, OUT2 and OUT3**[1] as arguments. Order can be random.

setFadeTime(float fadeValue = 0.0)

If fade function is utilised, user will have to set standard fade time. Argument is float type, because argument represents time in seconds. Available options are: **0.00, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 1.00, 2.00, 4.00, 6.00 and 8.00.** If no argument is provided, fade time will be set to 0.0.

setFadeType(bool fadeType = LINEAR, int firstOutput = false, int secondOutput = false, int thirdOutput = false, int fourthOutput = false[1])

If fade function is utilised, user can set if fade will be realised as linear function or exponential function. If no arguments are provided, all outputs will use linear function. For first argument user can use keywords: **LINEAR or EXPONENTIAL.** For other arguments, user can use keywords: **OUT0, OUT1, OUT2 and OUT3**[1]. Order can be random.

setCurrent(int OUT0CurrentValue = 0, int OUT1CurrentValue = 0, int OUT2CurrentValue = 0, int OUT3CurrentValue =0[1])

In order for channel to work correctly, user has to set the amount of current, that will flow via channel. Arguments can take values ranging from **0 to 255.** Value of 255 is the value of the maximum set current via **setMaxCurrent** function. Order of the arguments is not random, meaning that the first provided argument will be attached to first channel (OUT0), second provided argument will be attached to second channel (OUT1), third provided argument will be attached to third channel (OUT2) and fourth provided argument will be attached to first channel (OUT3)[1].

1. Only for LP5814 and LP5816

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setPWM(int OUT0PWMValue = 0, int OUT1PWMValue = 0, int OUT2PWMValue = 0, int OUT3PWMValue =0[2])

In order for channel to work correctly, user has to set width of pulse (PWM) value, that will modulate the channel. Arguments can take values ranging from **0 to 255**, where 255 is the value of full brightness. Order of the arguments is not random, meaning that the first provided argument will be attached to first channel (OUT0), second provided argument will be attached to second channel (OUT1), third provided argument will be attached to third channel (OUT2) and fourth provided argument will be attached to first channel (OUT3)[2].

setInstantBlinking[3](bool value = DISABLE)

If main MCU is not ready for establishment of I^2C communication, LP5815 offers instant blinking feature. If STAT pin is driven logic high and instant blinking is not disabled, OUT0 will start to blink according to specified pattern (LP5815 datasheet, p.19). In order to disable instant blinking feature, user has to pass **false** as an argument or **DISABLE** keyword. Enabling instant blinking happens on – start, but it is possible to enable this feature manually passing **true** as an argument or **ENABLE** keyword.

setOutEngine[4](int OUT0Engine = ENGINE0, int OUT1Engine = ENGINE1, int OUT2Engine = ENGINE2, int OUT3Engine = ENGINE3[5])

Unique engine modules are installed in LP5814 and LP5815 devices. These four engines allow user to pre – program animation pattern, that can be used by the device standalone, regardless of host MCU. Engine has to be assigned to specified output chosen by user. Single engine can be applied to multiple outputs at the same time. User can use keywords: **OUT0, OUT1, OUT2 and OUT3**[5] for patterns, **ENGINE0, ENGINE1, ENGINE2** and **ENGINE3** for selecting the engine. Order of the arguments is not random, meaning that the first provided argument will be attached to first channel (OUT0), second provided argument will be attached to second channel (OUT1), third provided argument will be attached to third channel (OUT2) and fourth provided argument will be attached to first channel (OUT3)[5].

setEnginePatternOrder[4](int engine = ENGINE0, int pattern0 = PATTERN0, int pattern1 = PATTERN1, int pattern2 = PATTERN2, int pattern3 = PATTERN3)

User can specify in what order patterns (determined by user) will be displayed by engine. Therefore, user can control which patterns will be controlled by specific output. Before that, patterns must be defined by the user. Order of the arguments is not random, meaning that the first provided argument will indicate to which engine patterns are assigned to, second

2. Only for LP5814 and LP5816
3. Only for LP5815
4. Only for LP5814 and LP5815
5. Only for LP5814

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provided argument will set first pattern, third provided argument will set second pattern, fourth provided argument will set third pattern and fifth provided argument will set fourth pattern. Single pattern can be assigned multiple times. User can use keywords: **PATTERN0, PATTERN1, PATTERN2** and **PATTERN3** for patterns, **ENGINE0, ENGINE1, ENGINE2** and **ENGINE3** for selecting the engine.

enableEnginePatternOrder[6](int engine = ENGINE0, int patternEnable0 = ENABLE, int patternEnable1 = ENABLE, int patternEnable2 = ENABLE, int patternEnable3 = ENABLE)

Even after setting pattern order, user must enable patterns. Additionally user can disable certain patterns, if it is necessary. To enable pattern, user has to provide **ENABLE** keyword, as an argument. In order to disable channel, user has to provide **DISABLE** keyword. To set proper behaviour on proper engine, user can use keywords: **ENGINE0, ENGINE1, ENGINE2** and **ENGINE3.**

SetEngineRepeatTime[6](int engine = ENGINE0, int times = 1)

Each engine can repeat displayed pattern specified amount of time, before ending its operation. To set amount of repeat times, user has to provide one of these keywords: **ONCE, TWICE, THRICE** and **INFINITE**. To set proper behaviour on proper engine, user can use keywords: **ENGINE0, ENGINE1, ENGINE2** and **ENGINE3.**

SetPatternPauseTime[6](int pattern = PATTERN0, float pauseT0 = 0.0, float pauseT1 = 0.0)

There is a possibility to set pause time between the end of pattern and the beginning of the pattern. It is important to note, that pause times is set for particular pattern. For more information regarding timing of a pattern, please see LP5815 datasheet, p.16. Second and third argument are float types, because arguments represent time in seconds. Available options are: **0.00, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 1.00, 2.00, 4.00, 6.00 and 8.00.** If no arguments are provided relating pause time, pause times will be set to 0.0. First argument decides for which pattern pause times are set. User can provide keywords for the first argument: **PATTERN0, PATTERN1, PATTERN2** and **PATTERN3.**

setPatternRepeatTime[6](int pattern = PATTERN0, int repeatTimes = 16)

Each pattern can be displayed specified amount of times in a row during engine cycle. Normally, pattern is set to run indefinitely, but user can change the value of repeat times by using second argument. Available options that set the amount of repeat cycles for pattern are: **1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15** and a keyword **INFINITE.** In order to choose specified pattern, user can provide keywords for the first argument: **PATTERN0, PATTERN1, PATTERN2** and **PATTERN3.**

6. Only for LP5814 and LP5815

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setPWMPattern[7](int pattern = PATTERN0, int firstPWM = 0, int secondPWM = 0, int thirdPWM = 0, int fourthPWM = 0, int fifthPWM = 0)

This function is responsible for setting the actual intensity of specified pattern. Pattern is divided into five slices, that can be configured individually. First argument specifies which pattern will be modified. User for this purpose can use specified keywords: **PATTERN0, PATTERN1, PATTERN2** and **PATTERN3.** Following five arguments can take values ranging from **0 to 255**, where 255 is the value of full brightness. For more information, please refer to LP5815 datasheet, p.16.

setPatternSloperTime[7](int pattern = PATTERN0, float firstTime = 0.0, float secondTime = 0,0, float thirdTime = 0.0, float fourthTime = 0.0)

Duration of specified PWM pattern slices can be modified. Arguments are float type, because argument represents time in seconds. Available options are: **0.0, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 1.00, 2.00, 4.00, 6.00 and 8.00.** If no argument is provided, duration of slice will be set to 0.0. First argument specifies which pattern will be modified. User for this purpose can use specified keywords: **PATTERN0, PATTERN1, PATTERN2** and **PATTERN3.** For more information about slice duration, please refer to LP5815 datasheet, p.16.

setWorkingMode[7](bool workingMode = MANUAL, int firstOutput = false, int secondOutput = false, int thirdOutput = false)

This function sets output working mode. There are two possibilities: **MANUAL** or **AUTOMATIC.** In manual mode, user can manually set output properties. In automatic mode, engines are responsible for output properties. First argument uses mentioned above keywords, next arguments assign mode to proper output. User can use keywords: **OUT0, OUT1, OUT2 and OUT3**[8] as arguments. Order can be random.

startAutonomousAnimation[7]()

If all engines and patterns are set by the user to their proper values, enabling the animation requires calling this command. Function does not utilise additional input arguments. Call this function after update() function.

pauseAutonomousAnimation[7]()

User can pause animation at any moment by calling this function. Function does not utilise additional input arguments.

continueAutonomousAnimation[7]()

User can continue animation, after pausing, at any moment by calling this function. Function does not utilise additional input arguments.

7. Only for LP5814 and LP5815
8. Only for LP5815

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stopAutonomousAnimation[9]()

User can stop entirely animation at any moment by calling this function. This can be useful if there is need to manually driver LED. Function does not utilise additional input arguments.

checkFlag(int flag)

Device can raise a flag in proper register, if it detects specified event. Purpose. Possible flags are: **Power On Reset (POR), Thermal ShutDown (TSD), Engine Busy[9], OUT0 Engine select[9], OUT1 Engine select[9], OUT2 Engine select[9] and OUT3 Engine select[10]**. In order to check the state of selected flag, user can use keywords: **POR_FLAG, TSD_FLAG, ENGINE_BUSY_FLAG[9], OUT0_ENGINE_BUSY_FLAG[9], OUT1_ENGINE_BUSY_FLAG[9], OUT2_ENGINE_BUSY_FLAG[9] and OUT3_ENGINE_BUSY_FLAG[10].** Function will return bool type value.

clearFlag(int flag)

User can clear two fault flags: **Power On Reset (POR)** and **Thermal ShutDown (TSD).** Clearing flags is mandatory for proper operation. In order to clear flag, user has to provide proper keyword: **POR_FLAG** or **TSD_FLAG.**

Keywords

User can use following keywords in designed sketch:

Selecting output: OUT0, OUT1, OUT2, OUT3
Fade type: EXPONENTIAL, LINEAR
Working mode: AUTOMATIC, MANUAL
Engine selection: ENGINE0, ENGINE1, ENGINE2, ENGINE3
Pattern selection: PATTERN0, PATTERN1, PATTERN2, PATTERN3
Permission: ENABLE, DISABLE
Repeat times: ONCE, TWICE, THRICE, INFINITE

9. Only for LP5814 and LP5815
10. Only for LP5814

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